

Nutrition & Fitness Tracker Module PRD

Overview

Build a Nutrition & Fitness Tracker integrated into the existing Firebase-based Expense Tracker. The module should support nutrition tracking, meal logging, goals, BMI, weight tracking, workout tracking, AI nutrition analysis, dashboards, reports, and historical analytics.

Tech Stack

Next.js/React, TypeScript, Tailwind CSS, Firebase Authentication, Cloud Firestore, Firebase Storage, Cloud Functions (optional).

Core Features

User profile, goals, dynamic daily meal frequency, meal logging, nutrition calculations, dashboards, progress tracking, workout tracking, water tracking, reports, notifications, AI assistant.

Dynamic Daily Meal Frequency

Meal frequency is independent for every day. Each day the user selects how many meals they plan to eat (1-6 or custom). Generate meal cards dynamically. Allow add/remove/rename/reorder meals. Previous days must never be affected.

Meal Entry

Each meal stores name, time, foods eaten, notes and optional photo. AI calculates calories, protein, carbohydrates, fat, fiber and micronutrients.

Dashboard

Daily totals, macro progress, BMI, calories, weight, water, charts, weekly/monthly trends.

Firebase

Collections: users, profile, goals, daily_logs, meals, nutritionSummary, weight_history, body_measurements, workout_history, reports, settings.

Implementation

Use reusable components, clean architecture, responsive UI, offline support, Firestore security rules, optimistic updates, error handling, production-ready code.